Matthew S. Scott

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Education

PhD in Mathematics, *University of British Columbia* 2023-2027 (expected)
Supervisors: Yaniv Plan, Özgür Yılmaz

MSc in Mathematics, *University of British Columbia* 2021-2023
Thesis: *Generative Compressed Sensing with Fourier Measurements*. Grade average 92%.

BSc, Joint Honours Mathematics and Physics, *McGill University* 2017-2021
First Class Honours; minor in Computer Science; GPA 3.89/4.00.

DEC, Honours Pure and Applied Science, *Marianopolis College*, Montreal 2015-2017

Publications

- Y. Plan, **M. S. Scott**, O. Yilmaz, *Exact signal recovery from generative priors with sigmoid output activation*, Sampling Theory, Signal Processing, and Data Analysis, submitted.
- **M. S. Scott**, *Convexity from order: a theory of faces and generalized separation*, J. Convex Analysis, submitted.
- Y. Plan, **M. S. Scott**, O. Yilmaz, *Partially deterministic sampling for compressed sensing with denoising guarantees*, Information and Inference (IMA), submitted.
- M. P. Friedlander, S. Kubal, Y. Plan, **M. S. Scott**, *Average-case thresholds for exact regularization of linear programs*, SIAM J. Optimization, second round of revisions.
- **M. S. Scott**, *Bilateral facial reduction: qualification-free subdifferential calculus and exact duality*, Set-Valued and Variational Analysis, major revisions.
- Y. Plan, **M. S. Scott**, X. Sheng, O. Yilmaz, *Denoising guarantees for optimized sampling schemes in compressed sensing*, SIAM J. Mathematics of Data Science, 2026.
- A. Berk, S. Brugiapaglia, Y. Plan, **M. S. Scott**, X. Sheng, O. Yilmaz, *Model-adapted Fourier sampling for generative compressed sensing*, NeurIPS 2023 Workshop on Deep Learning and Inverse Problems, oral (6 of 39).
- A. Berk, S. Brugiapaglia, B. Joshi, Y. Plan, **M. S. Scott**, O. Yilmaz, *A coherence parameter characterizing generative compressed sensing with Fourier measurements*, IEEE J. Selected Areas in Information Theory, 2022.

Experience

Instructor, Teaching and Research Assistant, *University of British Columbia* 2021-present

- Instructor of record: MATH 180 (Fall 2024), MATH 100 (Fall 2025), MATH 180 (Fall 2026).
- Research in generative compressed sensing, high-dimensional probability and convex analysis; numerics in Julia.
- Teaching assistant: grading and tutoring in the Math Learning Centre.

Freelance Tutor, *Superprof*, Montreal 2020-2021

- Rated 4.87/5 over 15 reviews. Earlier: math tutor at Paramount Study School (2017) and Collège Notre-Dame (2015).

Research Assistant in Computer Science, *McGill University (Prakash Panangaden)* Summer 2020

- Adapted Yaida's fluctuation-dissipation learning-rate method to actor-critic reinforcement learning (PyTorch); proved results and improved performance. Funded by an NSERC USRA.

Machine Learning Intern, *Decathlon Canada*, Montreal Summer 2018

- Built a recurrent-network sales forecaster that beat 4 of 5 in-house models, the only one predicting long term.
- Built a Python (Flask) API ranking sport popularity from Google search data.

Web Application Developer, *GoRush*, Montreal Summer 2017

- Sole developer of an on-demand lawn-mowing app (JavaScript, Meteor, MongoDB).

Software projects

- **Latex Exporter** (TypeScript, 5k lines): Obsidian plugin that compiles a graph of linked Markdown notes (embeds, citations, theorems, cross-references) into one LaTeX paper; 4,700+ downloads, 44 GitHub stars, test suite.
- **Research numerics** (Julia, Flux.jl; Python): generative priors (VAEs) and signal-recovery experiments for the compressed-sensing papers; batch pipelines on Slurm clusters. Packages GenerativeRecoveries.jl, VaeModels.jl.

- **Cracker Craft** (2016; C#, Unity, Blender): Android fireworks game, 10,000+ downloads on Google Play.
- **Machinations** (2021): turn-based multiplayer duelling game.
- **Line-following robot** (2015): PID control in C on an Arduino, with its electronics.
- Animated video explaining the Legendre transform (2022).

Awards

- UBC Graduate Departmental Award, 2021 and 2022.
- NSERC Undergraduate Student Research Award, 2020.
- Ernest Fox Award for Excellence in Mathematics, Marianopolis College, 2017 (one graduating student per year).
- Robohacks 2017: Best CEGEP and Most Imaginative Robot. McGameJam 2017: Special Mention.
- Canadian Computing Competition, top 25%, 2016 and 2017.
- Fermat Math Contest, school champion, 2015. AMQ Math Competition, 7th of 161, 2014.

Talks

- *Qualification-free duality via reduction to the joint supporting subspace*, West Coast Optimization Meeting, UW Tacoma, 2026.
- *Denoising guarantees for optimized sampling*, 5th Biennial SIAM Pacific Northwest Meeting, UW Seattle, 2025.
- *Exact regularization of linear programs*, 5th Biennial SIAM Pacific Northwest Meeting, UW Seattle, 2025.
- *Model-adapted Fourier sampling for generative compressed sensing*, NeurIPS Workshop on Deep Learning and Inverse Problems, New Orleans, 2023 (oral).
- *Generative recovery from subsampled Fourier measurements*, CMS Winter Meeting, Montreal, 2023; UBC MILD seminar, 2022.
- Group talks: terminal embeddings (2022-23); convex sets and faces, optimal transport and conic duality (2026).

Service

- Referee, Journal of Optimization Theory and Applications (2026).
- Workshop on Obsidian and Zettelkasten note-taking for research, UBC Institute of Applied Mathematics (talk 2022, workshop 2024).

Skills

- **Mathematics:** high-dimensional probability and statistics, random matrices, convex optimization and duality, compressed sensing, geometry and expressivity of neural networks.
- **Programming:** Python (PyTorch, NumPy, Flask), Julia (Flux.jl), TypeScript/JavaScript, C#, Java; Git, Linux, Slurm, LaTeX, Typst.

Interests

- Singing (choirs, an a cappella group), acting (the musical *First Date*), Jiu-Jitsu.